

1. Which of the following best defines encapsulation?

- a) Wrapping data and methods into a single unit
- b) Using multiple classes for the same function
- c) Creating objects from classes
- d) Hiding the implementation details from the user

2. In Java, encapsulation is achieved mainly using:

- a) abstract classes
- b) private fields with getters and setters
- c) final keyword
- d) Interfaces

3. What is the main advantage of encapsulation?

- a) Faster code execution
- b) Code reusability
- c) Data hiding and security
- d) Supports multiple inheritance

4. Which access modifier provides the highest level of encapsulation?

- a) public
- b) private
- c) protected
- d) default

5. In encapsulation, how do we usually allow controlled access to private fields?

- a) By making fields public
- b) By using constructors only
- c) By using getter and setter methods
- d) By using static variables

6. Which OOP concept allows a class to acquire properties of another class?

- a) Polymorphism
- b) Encapsulation
- c) Inheritance
- d) Abstraction

7. In Java, which keyword is used to inherit a class?

- a) super
- b) extends
- c) implements
- d) inherits

8. Which of the following is not a type of inheritance in Java?

- a) Single inheritance
- b) Multilevel inheritance
- c) Multiple inheritance through classes
- d) Multiple inheritance through interfaces

9. What is the base class of all classes in Java?

- a) Class
- b) Object
- c) Base
- d) Abstract

10. Which of the following statements about inheritance is correct?

- a) A subclass can inherit private members of the parent.
- b) A subclass cannot override methods of the parent.
- c) A subclass can access protected members of the parent.
- d) A subclass cannot have its own constructors.

11. Polymorphism means:

- a) Many methods with the same name but different parameters or behavior
- b) Wrapping code into objects
- c) Restricting access to class members
- d) Creating abstract classes

12. Compile-time polymorphism in Java is achieved through:

- a) Method overriding
- b) Method overloading
- c) Interfaces
- d) Abstract classes

13. Runtime polymorphism is implemented using:

- a) Method overloading
- b) Method overriding
- c) Constructor chaining
- d) Static binding

14. Which keyword is used to call the parent class method in case of overriding?

- a) parent
- b) this
- c) super

- d) base

15. Which of these allows polymorphic behavior in Java?

- a) Abstract classes and interfaces
- b) Encapsulation
- c) Static variables
- d) Packages

16. Abstraction focuses on:

- a) Hiding implementation details and showing only functionality
- b) Code reusability
- c) Creating multiple methods with the same name
- d) Security of data

17. In Java, abstraction is achieved using:

- a) Abstract classes and interfaces
- b) Final classes
- c) Static methods
- d) Constructors

18. Which of the following statements is true?

- a) Abstract class can have both abstract and non-abstract methods
- b) Interface can contain constructors
- c) Abstract methods can have a body
- d) Abstract classes cannot have instance variables

19. If a class has at least one abstract method, the class must be declared as:

- a) final
- b) abstract
- c) interface
- d) protected

20. Which Java feature was introduced to provide more power to abstraction in interfaces (from Java 8 onwards)?

- a) Multiple constructors in interfaces
- b) Default and static methods in interfaces
- c) Private constructors in interfaces
- d) Inheritance for multiple classes